

Siraj TM

BSMS Data Science, 5TH year
IISER TVM

CONTACT

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EXPERIENCE

MAJOR THESIS — IISER TVM

Guide - Dr. Raji Susan Mathew
August 2025 - Present

Phase -1 : Developed an interpretable voxel-wise deep reinforcement learning based model for Quantitative susceptibility mapping (QSM) reconstruction.
Phase 2 : Integrated model based unrolling within iterative deep reinforcement learning for improved data consistency in QSM reconstruction

INTERNSHIP - IISER-TVM

Guide - Dr. Raji Susan Mathew
Collaborated with Dr. Navchetan Awasthi, University of Amsterdam
December 2024 - August 2025

Worked on improving generalizability in QSM reconstruction, by incorporating contextual input-target pairs to DeepQSM, an existing benchmark model. The resulting work has been accepted at ICVGIP'25.

INTERNSHIP - IISER-TVM

Guide - Dr. Raji Susan Mathew and Dr. Reji
November 2024 - July 2025

Worked on remote sensing change captioning using state space models by incorporating change detection using graph neural networks

Internship - IISER TVM

Guide - Dr. Chandrakala Meena
Collaborated with Dr. Sandeep Chowdhary, IIASA, Laxenburg, Austria
August 2023 - July 2025

Analysed faculty hiring networks between India and the rest of the world with the aim of measuring the impact of different academic migration trajectories on careers

PUBLICATIONS

Accepted:

T. Apotheker, **Siraj T. M**, R. S. Mathew, and N. Awasthi. "Context-qsmnet: Enhancing Generalization in Quantitative Susceptibility Mapping with Context-Aware Deep Neural Networks." **Accepted at the Indian Conference on Computer Vision, Graphics, and Image Processing (ICVGIP), 2025.**

Siraj T. M, Reji V., R. S. Mathew. "Graph-Guided Change Detection and State Space Model Captioning with Semantic Uncertainty Estimation " **Accepted at MIGARS 2026.**

Under Review:

Siraj T. M, Meenkashi S., Reji V., R. S. Mathew. RL-QSM: An Interpretable Reinforcement Learning-Based Reconstruction for Quantitative Susceptibility Mapping. **Submitted to MICCAI 2026.**

Siraj T. M., Hari Krishnan S, Mathew Vincent, Sandeep Chowdhary and Chandrakala Meena. "Scientific mobility patterns of Indian researchers: Impact on career growth." **Submitted to Quantitative Science Studies (QSS)**, 2025. Available on arxiv

EDUCATION

INDIAN INSTITUTE OF SCIENCE, EDUCATION AND RESEARCH,
THRIUVANANTHAPURAM
BSMS, I2 Data Science
2021 - Present
CGPA: 8.00/10
Major CGPA: 8.4/10
TOEFL : 112/120

Bhavans Varuna Vidhyalaya, Thrikkakkara, Kochi
AISSE (2019) and AISSCE (2021) - **CBSE AISSCE: 96.8 %**

SKILLS

- QSM / MRI Reconstruction, Medical Imaging, Computer Vision
 - Python, C, MATLAB, C++, R, HTML, CSS, DBMS, JavaScript
 - Pytorch, Tensorflow, OpenCV, CUDA, MPI, OpenMP
 - Git, Bash, Linux,
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AWARDS

INSPIRE Scholarship awarded by DST-GOI Awarded to the top 1% of students based on Class 12 board exam performance. Scholarship for pursuing Natural and Basic Sciences at the BSMS level.

Conferences & Symposiums

- **ISMRM India Chapter 2026** - e-Poster (received certificate of merit)
 - **Frontier Symposium In Data Science (FS-DS) IISER TVM 2026** - Oral Presentation First runner up.
 - **ICVGIP 2025** - Poster Presentation
 - **FS-DS IISER TVM 2025** - Oral Presentation
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COURSES

AI and Medical Imaging | Artificial Intelligence | Machine Learning I & II |
Data Structures | Mathematical Statistics | Parallel and Distributed
Computing | High Performance Computing | Linear Algebra | Optimization

REFERENCES

Dr. Raji Susan Mathew
Master's Thesis Guide
Assistant Professor, Department of Data Science
IISER Thiruvananthapuram, Kerala, India
rajisusanmathew@iisertvm.ac.in

Dr. Navchetan Awasthi
Internship Guide
Assistant Professor, School of Artificial
Intelligence and Data Science
IIT Jodhpur, Rajasthan, India.
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Dr. Alwin Poulose
Course Instructor
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